

Foundations of Cybersecurity

Google Cybersecurity Professional Certificate

Week 2 – The Role of a Security Analyst

Understanding the Security Analyst Role

Security Analysts are responsible for protecting an organization's digital infrastructure. They monitor systems, identify suspicious behaviour, investigate alerts, and respond to incidents before they become major security breaches. Their work helps ensure the confidentiality, integrity, and availability of business data.

Key Responsibilities

- Monitor security alerts from different tools.
- Investigate suspicious activities and determine their impact.
- Protect computers, servers, cloud resources, and networks.
- Conduct security audits and vulnerability assessments.
- Work with IT teams to implement stronger security controls.
- Maintain documentation and create incident reports.

Daily Workflow

A typical day begins with reviewing alerts generated by monitoring tools. Analysts examine logs, identify false positives, investigate real threats, and document every action taken. They also participate in meetings, improve detection rules, and update security documentation.

Playbooks

A playbook is a documented guide that explains how to respond to a specific cybersecurity incident. For example, if a phishing email is detected, the playbook outlines each step—from isolating the affected account to notifying users and documenting the incident. Playbooks improve consistency and reduce response time.

Technical Skills

Successful analysts should understand:

- Networking fundamentals
- Linux and Windows administration
- SIEM platforms for log analysis
- Intrusion Detection Systems (IDS)

- Python and SQL basics
- Incident Response and Digital Forensics

Transferable Skills

Beyond technical knowledge, security analysts require strong communication, critical thinking, teamwork, curiosity, adaptability, and problem-solving abilities. These skills help analysts explain technical issues clearly and collaborate effectively during incidents.

Real-World Example

Imagine an employee clicks a malicious email attachment. The SIEM detects unusual activity and generates an alert. The analyst investigates the logs, isolates the compromised device, resets user credentials, removes malicious files, documents the incident, and recommends security awareness training to prevent similar attacks.

Key Takeaways

- Cybersecurity is both technical and analytical.
- Documentation is as important as investigation.
- Continuous learning is essential because cyber threats evolve constantly.
- Security analysts protect organizations through monitoring, investigation, response, and collaboration.